

1 Introduction

Consider the action that $SL_2(\mathbb{Z})$ has on $\mathbb{C}$ with $g \in SL_2(\mathbb{Z})$ giving $g(z) = \frac{az+b}{cz+d}$. From a calculation, it follows that $Im(g(z)) = \frac{Im(z)}{|cz+d|^2}$. Thus we conclude that the upper-half plane $\mathbb{H}$ is invariant under the action of $SL_2(\mathbb{Z})$ via the mentioned fractional linear transformation. Notice that the matrices

$$\begin{pmatrix} 1 & 0 \\ 0 & 1 \end{pmatrix}, \begin{pmatrix} -1 & 0 \\ 0 & -1 \end{pmatrix}$$

both act trivially on $\mathbb{H}$. In general, the action of $g \in SL_2(\mathbb{Z})$ is the same as that of $-g$, so it is more convenient to examine the *modular group*, $G := SL_2(\mathbb{Z})/\{\pm 1\}$. Set now

$$S := \begin{pmatrix} 0 & -1 \\ 1 & 0 \end{pmatrix}, T := \begin{pmatrix} 1 & 1 \\ 0 & 1 \end{pmatrix}$$

and it may be shown that S, T generate G . Let $D = \{z \in \mathbb{H} \mid |z| \geq 1, Re(z) \leq 1/2\}$.

Theorem.

1. For every $z \in \mathbb{H}$, there exists $g \in G$ so that $g(z) \in D$.
2. Let $z, z' \in D$ be two distinct points that are congruent modulo G . Then $Re(z) = \pm 1/2$ and $z = z' \pm 1$, or $|z| = 1$ and $z' = -1/z$.
3. Let $z \in D$ and $I(z) = \{g \in G \mid g(z) = z\}$, the stabilizer of z . Then $I(z) = \{1\}$, except in the following cases:
 - (a) $z = i$, in which $I(z) \cong \mathbb{Z}/2\mathbb{Z}$, generated by S .
 - (b) $z = e^{2\pi i/3}$, in which case $I(z) \cong \mathbb{Z}/3\mathbb{Z}$, generated by ST .
 - (c) $z = e^{\pi i/3}$, in which case $I(z) \cong \mathbb{Z}/3\mathbb{Z}$, generated by TS .

Corollary. The canonical map $\phi : D \rightarrow \mathbb{H}/G$ is surjective and $\phi|_{int(D)}$ is injective.

Proof. $\phi : D \rightarrow \mathbb{H}/G$ is given by sending $z \in D \mapsto [z] \in \mathbb{H}/G$. Suppose $[w] \in \mathbb{H}/G$. Clearly $[w]$ is nonempty, so let $a \in [w]$, and thus there is a $g \in G$ with $g(a) = w$. We may let $g = \text{id}$ and it follows trivially that ϕ is surjective. If instead $\phi|_{int(D)}$ was not injective, then there exists z and w distinct in $\mathbb{H}$ so that $[z] = [w]$. Part two of the previous theorem gives that since $z \equiv w \pmod{G}$, thus either $Re(z) = \pm 1/2$ or $|z| = 1$, but both cases lead to contradictions as we assumed $z \in int(D)$. $\square$

Let now $k \in \mathbb{Z}$. We say a function f is called *weakly modular of weight $2k$* if f is meromorphic on $\mathbb{H}$ and obeys

$$f(z) = (cz + d)^{-2k} \cdot f\left(\frac{az + b}{cz + d}\right)$$

for all matrices

$$\begin{pmatrix} a & b \\ c & d \end{pmatrix} \in SL_2(\mathbb{Z})$$

A weakly modular function is called simply *modular* if it is meromorphic at infinity. A modular function which is holomorphic on $\mathbb{H}$ (including at infinity) is called a *modular form*, but if it is 0 at infinity, we call it a *cusp form*. From a quick calculation, it follows that the product of two modular forms (defined pointwise on $\mathbb{H}$) is a modular with weight the sum of the weights.

Let Γ be a lattice in $\mathbb{C}$. The series $\sum'_{\gamma \in \Gamma} \frac{1}{|\gamma|^\sigma}$ is convergent for $\sigma > 2$, where $\sum'$ denotes the sum being taken over nonzero elements of the lattice. We get a similar notion of convergence depending on the exponent in the denominator of the sum, as seen in p -series in calculus. If $k \in \mathbb{Z}$, then $G_k(\Gamma) = \sum'_{\gamma \in \Gamma} \frac{1}{|\gamma|^{2k}}$ is modular of weight $2k$ and is called the *Eisenstein series* of index k . More generally, we will use the formula $G_k = G_k(z) = \sum_{(m,n) \neq (0,0)} \frac{1}{(mz+n)^{2k}}$, where m, n are integers.

If $k \in \mathbb{Z}$, let M_k denote the complex vector space of modular forms of weight $2k$, and let M_k^0 be defined similarly for the space of cusp forms. By definition of a cusp form, $M_k^0 = \ker(f \mapsto f(\infty))$, where $f \in M_k$. Thus we obtain that over $\mathbb{C}$, $\dim(M_k/M_k^0) \leq 1$, as we look at the space of scalars. It may be shown that for $k \geq 2$, $G_k \in M_k$ satisfies $G_k(\infty) \neq 0$, and so we obtain $M_k = M_k^0 \oplus \mathbb{C}G_k$. We now have the following theorem.

Theorem. The space M_k has for basis the family of monomials $G_2^\alpha G_3^\beta$, with $\alpha, \beta \in \mathbb{Z}_{\geq 0}$ and $2\alpha + 3\beta = k$.

Thus we deduce that for a fixed k , the space of modular forms M_k is finite-dimensional, as there are only finitely many solutions in positive integers α, β to the equation $2\alpha + 3\beta = k$. Now put $M := \bigoplus_{k=0}^{\infty} M_k$ and define a homomorphism $\phi : \mathbb{C}[x, y] \rightarrow M$ so that $x \xrightarrow{\phi} G_2$ and $y \xrightarrow{\phi} G_3$, and it follows that the previous theorem is equivalent to the statement that ϕ is an isomorphism. Thus we may naturally view M as a polynomial algebra $\mathbb{C}[G_2, G_3]$.

2 Hecke operators

We now give a few definitions. For E a set and X_E the free abelian group generated by E , we define a *correspondence* on E to be a homomorphism $T : X_E \rightarrow X_E$ satisfying the following property: $T(x) = \sum_{y \in E} n_y(x)$ (for $x \in E$, n_y an almost-zero integral function on E). For $\mathcal{R}$ the set of lattices in $\mathbb{C}$ and $n \in \mathbb{Z}_{\geq 1}$, define the function $T(n)$ (a *Hecke operator*) on $\mathcal{R}$ so that

$$T(n)\Gamma = \sum_{[\Gamma:\Gamma']=n} \Gamma'$$

For our purposes, we define a Hecke operator with respect to its action on a modular form. Suppose now f is a modular form with q -expansion $\sum c(m)q^m$. We define its

Hecke operator to be

$$T(n) = \sum f\left(\frac{a\tau + b}{d}\right), \quad \begin{pmatrix} a & b \\ 0 & d \end{pmatrix} \in SL_2(\mathbb{Z}), \quad ad = n, \quad 0 \leq b < d$$

More explicitly, we have

$$T(n)f = \sum_m \left[\left(\sum_{d|(m,n)} d^{k-1} \cdot c\left(\frac{mn}{d^2}\right) \right) q^m \right]$$

As an example, consider the Hecke operator $T(2)$. Suppose f has q -expansion $c(0) + c(1)q + c(2)q^2 + \dots$. Then for $m = 0$, the 0th coefficient of $T(2)f$ is equal to $\sum_{d|2} d^{k-1} \cdot c(0/d^2) = c(0) + 2^{k-1}c(0)$. Similarly, the 1st and 2nd coefficients of $T(2)f$ are $c(2)$ and $c(4) + 2^{k-1}c(1)$, respectively, and the other coefficients may be calculated in a similar fashion.

Now we give an example regarding the Ramanujan τ function. Take the discriminant cusp form (of weight 12), namely

$$\Delta = q \cdot \prod_{n>0} (1 - q^n)^{24}$$

Expanding the first several terms of Δ gives $q - 24q^2 + 252q^3 - 1472q^4 + 4830q^5 - 6048q^6 + \dots$. Note that the m th coefficient (for $m \geq 0$) of Δ is $\tau(m)$. It may be shown that the eigenvalue of $T(n)$ on Δ is $\tau(n)$, where τ is the Ramanujan τ function. We quickly check the cases $n = 2, 3$. In the first case, the relation of the q^1 -terms holds trivially. $T_2\Delta$ has q^2 -coefficient $\tau(4) + 2^{11} \cdot \tau(1)$ (obtained from the definition of a Hecke operator applied to a modular form) which is precisely to $\tau(2) \cdot (-24)$, where -24 is the q^2 -coefficient of Δ . $T_2\Delta$ has q^3 -coefficient $\tau(6)$, which is equal to $\tau(2) \cdot 252$, where 252 is the q^3 -coefficient of Δ . A similar calculation using the Hecke operator definition shows the case for $n = 3, 4, \dots$.

3 Weakly holomorphic modular forms

A *weakly holomorphic modular form* of weight $k \in 2\mathbb{Z}$ (for $\Gamma = PSL_2(\mathbb{Z})$) is a holomorphic function f on the upper-half plane $\mathbb{H}$ that satisfies

$$f\left(\frac{a\tau + b}{c\tau + d}\right) = (c\tau + d)^k f(\tau) \quad \begin{pmatrix} a & b \\ c & d \end{pmatrix} \in \Gamma$$

and which has a q -expansion of the form $f(\tau) = \sum_{m \geq n_0} a(m)q^m$, where $q = \exp(2\pi i\tau)$ and $n_0 = \text{ord}_\infty(f)$ (where ord is defined in the sense of a Laurent series). For notation, we write $\mathcal{M}_k$ to denote the space of all weakly holomorphic modular forms of weight k . As will become clear soon, we also define the weight 12 cusp form and the weight 0 modular function (j -function), which are, respectively,

$$\Delta(\tau) = q \prod_{n \geq 1} (1 - q^n)^{24} = \sum_{n \geq 1} \tau(n)q^n$$

and

$$j(\tau) = \frac{E_4(\tau)^3}{\Delta(\tau)} = q^{-1} + 744 + \sum_{n \geq 1} c(n)q^n$$

Now let $\mathcal{F}$ be the fundamental domain for $\Gamma = PSL_2(\mathbb{Z})$. Then, looking at the modular forms of weight k , we may write $k = 12l + k'$, where l is determined uniquely and k' is one of 0, 4, 6, 8, 10, 14. Then for each integer $m \geq -l$, there exists a unique $f_{k,m} \in \mathcal{M}_k$ so that $f_{k,m}(\tau) = q^{-m} +$ (higher-order terms). Then using the weight 12 cusp form and the Eisenstein series $E_{k'}$, we may write $f_{k,m} = \Delta^l E_{k'} F_{k,D}(j)$, where $F_{k,D}(x)$ is a polynomial in x of degree $D = l + m$ with integral coefficients. From this we get a basis of $\mathcal{M}_k$ by the collection of all forms $f_{k,m}$ for $m \geq -l$, with l defined in the equation $k = 12l + k'$. We can also write

$$f_{k,m}(\tau) = q^{-m} + \sum_n a_k(m, n)q^n$$

An important result of Siegel's is that if $k > 0$ then the coefficient $a_k(0, l + 1)$ is divisible by every prime p with $(p - 1) \mid k$. We look at the following problem:

Problem: What are the divisors of $a_k(m, n)$? Are there interesting patterns of the coefficients $a_k(m, n)$ for varying values of k, m, n ?

We now revisit the Ramanujan tau function. Lehmer's conjecture states that $\tau(n) \neq 0$ (for $n \geq 1$) iff in $f_{-10,n}(\tau) = q^{-n} - \tau(n)q^{-1} + \dots$, the initial term q^{-n} does not vanish. We may also look at the following problem:

Problem: What are other such non-vanishing results for $f_{k,n}$ for k not necessarily 10?